

Isabela Sforcin

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EDUCATION

University of California, Los Angeles (UCLA)

Master of Science: Computer Science (Incoming)

Incoming March 2027

University of California, Riverside

Bachelor of Science: Computer Engineering

Expected December 2026

Relevant Coursework: Robotics, Machine Learning, Artificial Intelligence, Database Systems, Statistics.

RESEARCH FELLOWSHIPS & GRANTS

- **Fellowships:** Awarded competitive federal/national research placements: NSF REU (Data Mining & Machine Learning), USDA Digital Agriculture Fellowship, and CRA DREU Fellowship (USC Interaction Lab).
- **NSF ACCESS Grant:** Co-PI and Allocation Manager on a successful NSF ACCESS grant, securing and managing high-performance computing resources for robot learning research at the GLAMOR Lab under Dr. Jesse Thomason.

RESEARCH EXPERIENCE

CRA DREU Fellow

Interaction Lab, University of Southern California (Advisor: Dr. Maja Matarić)

June 2026 – Present

Los Angeles, CA

- Designed and built a human-robot interaction study measuring how commitment tools affect user engagement, integrating embedded systems, Python, and cloud computing into a working end-to-end system.
- Applied FaceNet facial embeddings and gaze/eye-tracking, using cosine similarity between embedding vectors to classify participant engagement level; analyzed results in SQL to surface behavioral trends.

Research Assistant

GLAMOR Lab, University of Southern California (Advisor: Dr. Jesse Thomason)

October 2025 – June 2026

Los Angeles, CA

- Adapted a transformer-based Vision-Language-Action (VLA) model (OG-VLA) for bimanual robotic manipulation using GENIMA methods, generating high-fidelity spatial heatmap representations to map visual embeddings to action predictions.
- Resolved critical data-type inconsistencies (uint8-to-NumPy transformations) within a Diffusion Model preprocessing pipeline, restoring accurate loss calculation and enabling model convergence.
- Applied Tucker Decomposition as a dimensionality-reduction technique on large-scale robotics model representations, integrated with mechanistic-interpretability-driven fine-tuning of latent features.
- Conducted end-to-end evaluation on physical robotic hardware, using real-world trajectory data to iteratively refine model weights and performance.
- Served as Co-PI and Allocation Manager for a successful NSF ACCESS grant, managing HPC resources for lab-wide model training and experimentation.

NSF REU Fellow — Data Mining & Machine Learning

University of California, Riverside (Advisor: Dr. Vagelis Papalexakis)

June 2025 – October 2025

Riverside, CA

- Built a dataset-distillation pipeline combining Tucker Decomposition (dimensionality reduction), k-means clustering (latent clustering), and gradient matching in PyTorch/CUDA, reaching 96% downstream classifier accuracy using roughly 0.01% of the original training data.
- Benchmarked distilled latent representations against full training sets and presented findings in a written research summary submitted at the program's conclusion.

USDA Digital Agriculture Fellow

University of California, Riverside (Advisor: Dr. Zaowei Tan)

June 2024 – July 2025

Riverside, CA

- Architected a distributed data pipeline integrating a Raspberry Pi with SX1262 LoRa modules to ingest and decode remote sensor data for cloud-based analysis.

Researcher & Systems Engineer

Formula SAE Autonomous Vehicles, UC Riverside

January 2024 – Present

Riverside, CA

- Developed real-time embedded firmware (C/C++, STM32CubeIDE) bridging ROS perception nodes with vehicle actuators via CAN bus; performed Hardware-in-the-Loop validation prior to deployment.

Software Engineer (Part-Time)

U.S. Metals

2020 – Present

Riverside, CA

- Progressed from IT Intern to Software Engineer over six years while completing a full-time engineering degree; designed a computer vision pipeline (OpenCV) that automatically tracks and classifies vehicle entry/exit and recognizes returning customers via a barcode-based identification system, cutting per-transaction ticketing time from three minutes to forty seconds.

TECHNICAL SKILLS

Programming: Python (Expert), C/C++, CUDA, SQL, Bash, LaTeX

Deep Learning & Architectures: PyTorch, CNNs, transformer-based architectures, Vision-Language-Action (VLA) models, Diffusion Models

Embeddings & Representation Learning: FaceNet, facial/pose embeddings, cosine similarity, latent clustering (k-means), Tucker Decomposition and other dimensionality-reduction techniques, dataset distillation

Vision & Recognition: OpenCV, facial recognition, gaze/eye-tracking, behavioral and engagement inference

Robotics & Embedded Systems: ROS, bimanual manipulation, real-time vision-to-action integration, STM32CubeIDE, Raspberry Pi, HIL validation

Systems & Tooling: HPC resource management (NSF ACCESS), Git/GitHub, Docker, Linux development environments

AFFILIATIONS & AWARDS

IEEE Vice President | Society of Hispanic Professional Engineers (SHPE)